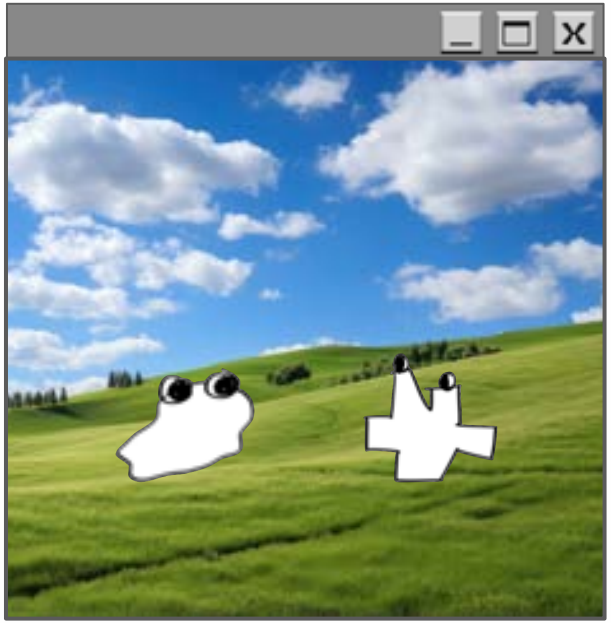
pebble.

prototype testing

Team 4



Assignment 5
cs 147 autumn 2026

Team Members



Ryan L.

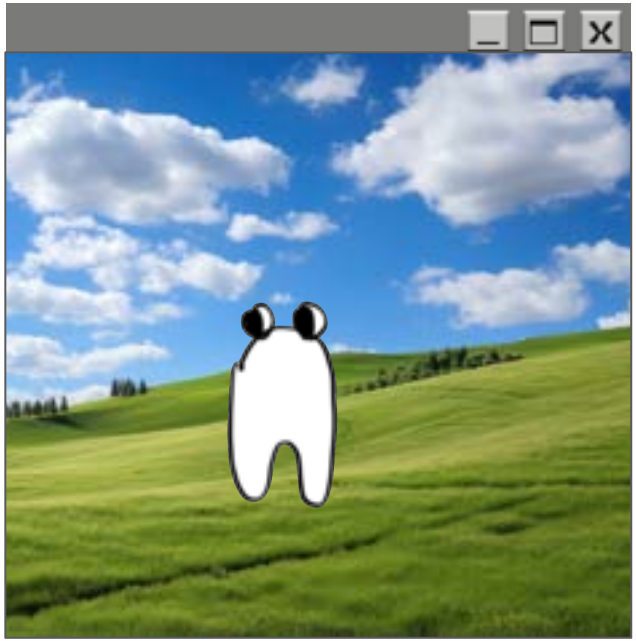


Brian L.



Cherie C.

SLIDESGO



pebble.

imperfection rocks

- We want users to slowly accumulate **imperfect** drawings until they create a greater artifact (their gallery) that is sharable with other users
- The value proposition has a double meaning: we call user drawings “pebbles”, and since they are not perfect, slogan is **“imperfection rocks”**



pebble.

imperfection rocks

→ Daily timed drawing prompts and platform for **busy** students and professionals with a hunger for creativity but no time

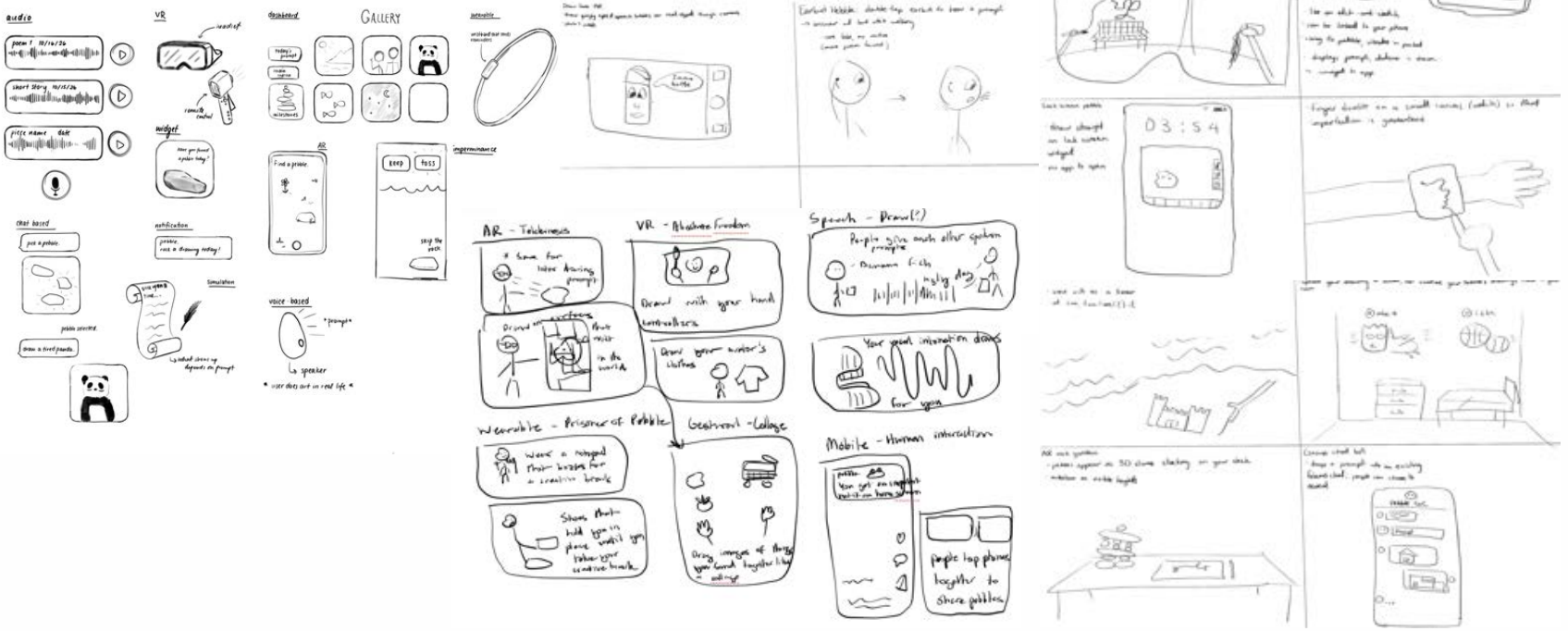
→ See **other users' drawings** for the same prompt after completion, invite friends, and reach milestones

01 Sketches

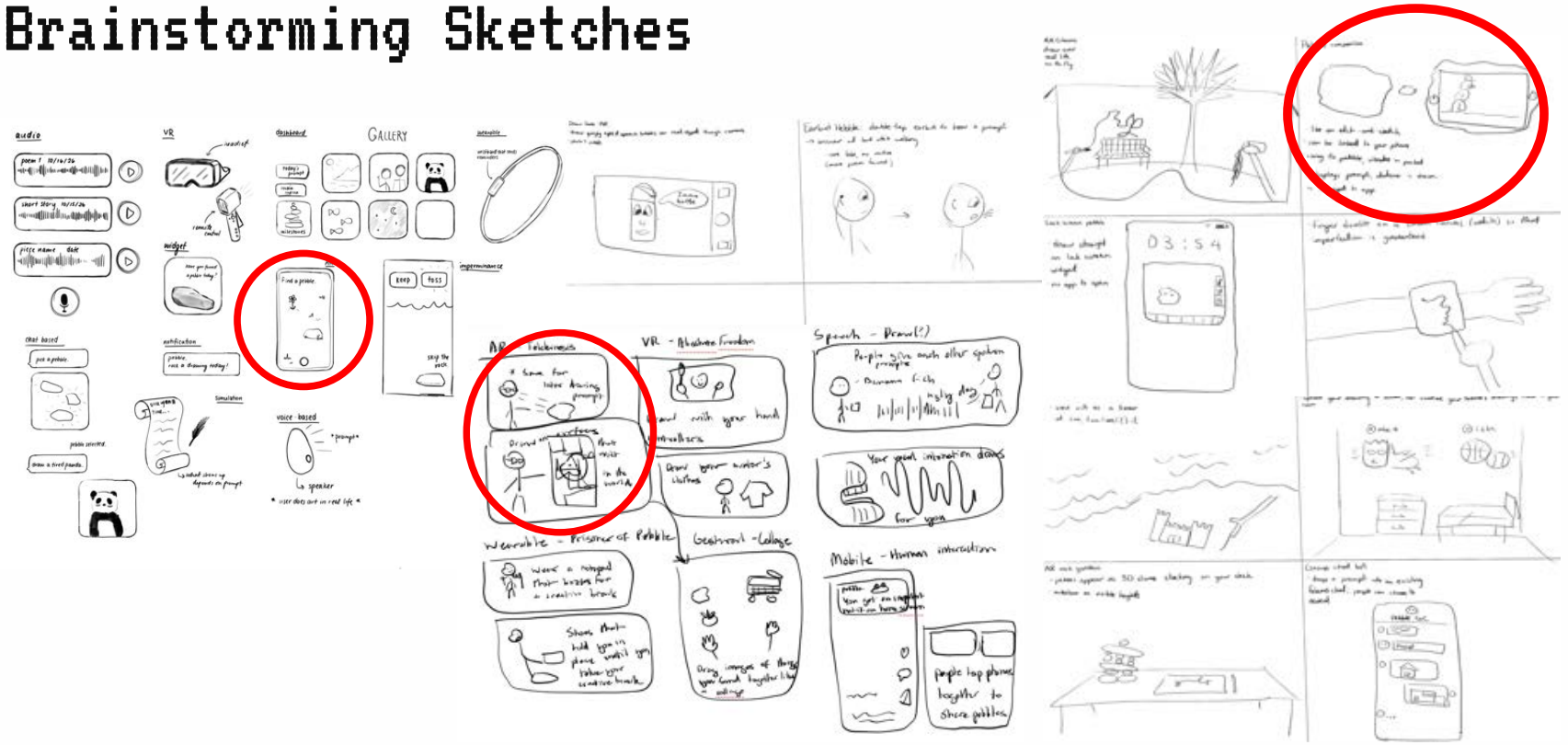


Brainstorming, Concept, Our Fleshed-Out Realizations

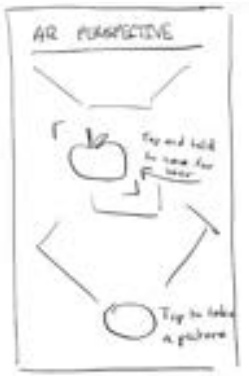
Brainstorming Sketches



Brainstorming Sketches



Concept Sketches - AR Realization



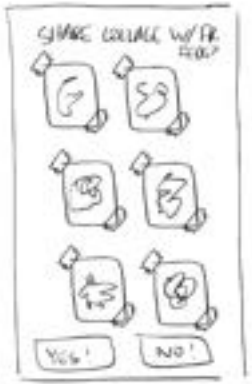
User goes about their day and finds object of interest



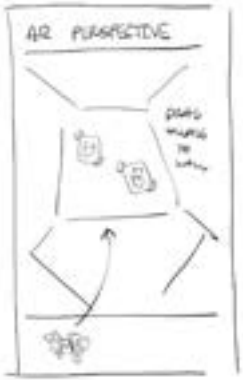
User can look at their previously shared stickers to gain inspiration/remember an intentionally important object.



User gets a notification on their phone that it's time to build their pebble. They can use previously scanned (?) stick and draw in relation to them.



User can choose to share their sketch/village with their friends (event expose)



User can place memories on geo-anchored rooms locations

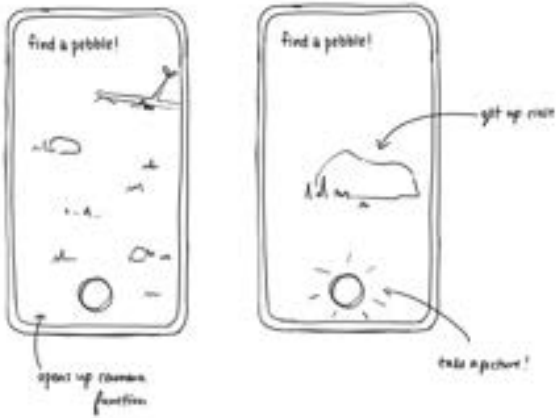


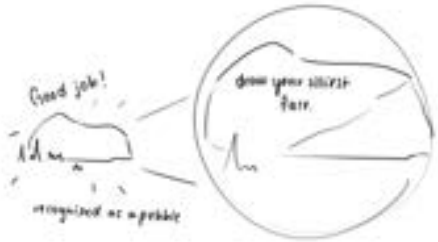
User can flip through all their villages

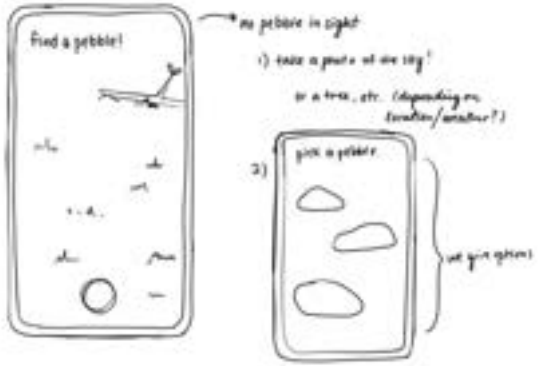
Concept Sketches - Physical Product Realization



Concept Sketches - Camera Work Realization



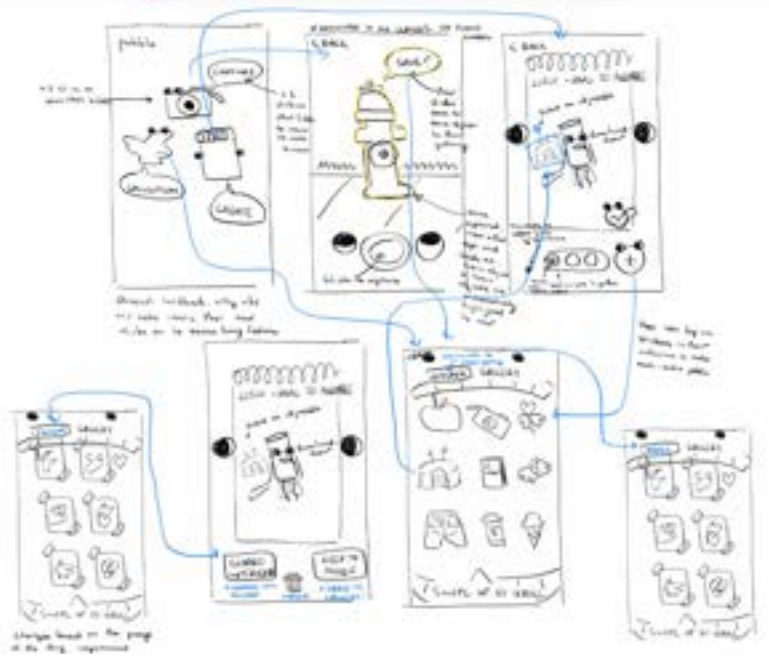




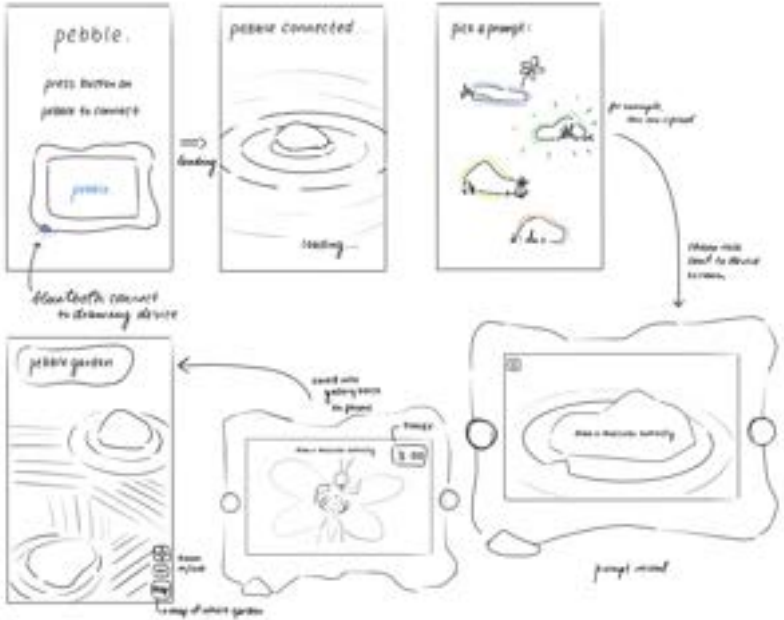
Our Focused Realizations

Our 2 Focused Realizations

01

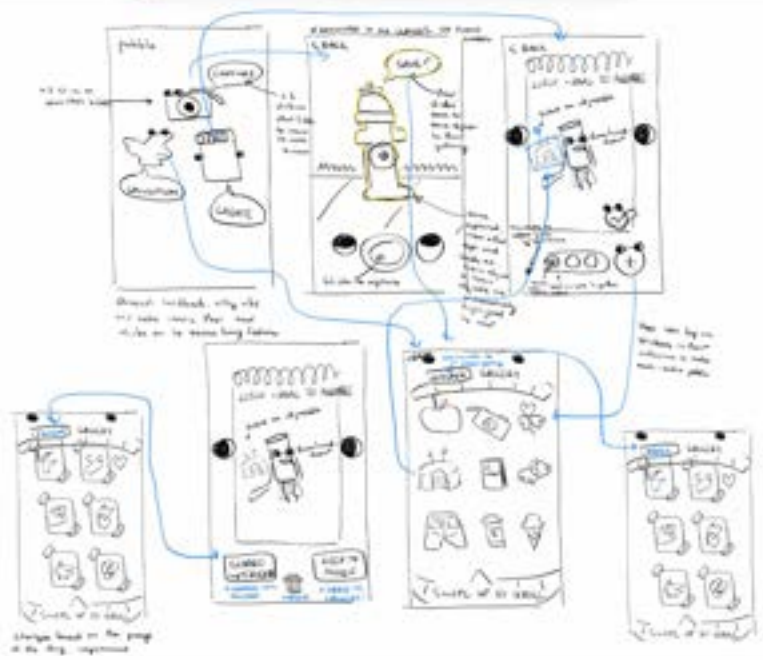


02



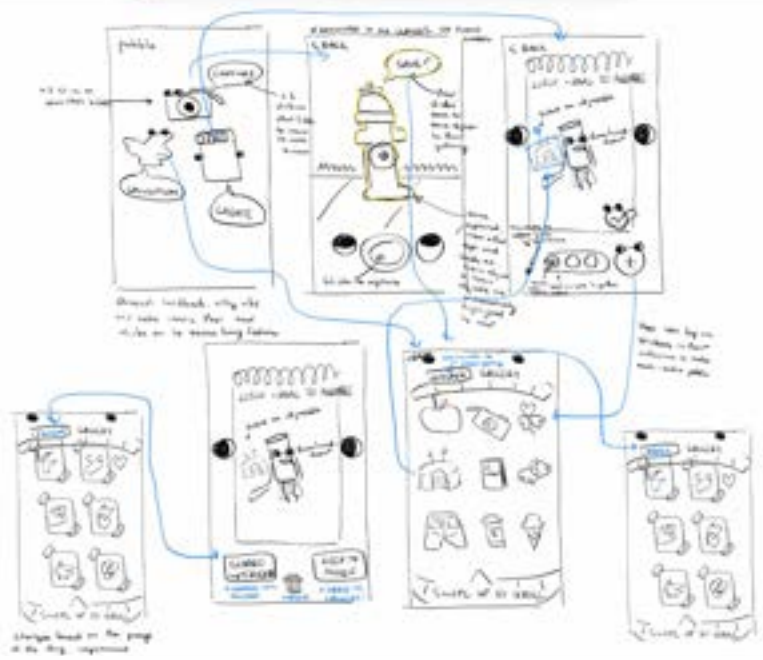

Selected
Interface
and Rationale
OK
Cancel
Apply

AR Sticker Collage: Pros



1. Interactive with the real world
2. Low barrier to entry
3. Reusable stickers
4. Allow easy share options with friends

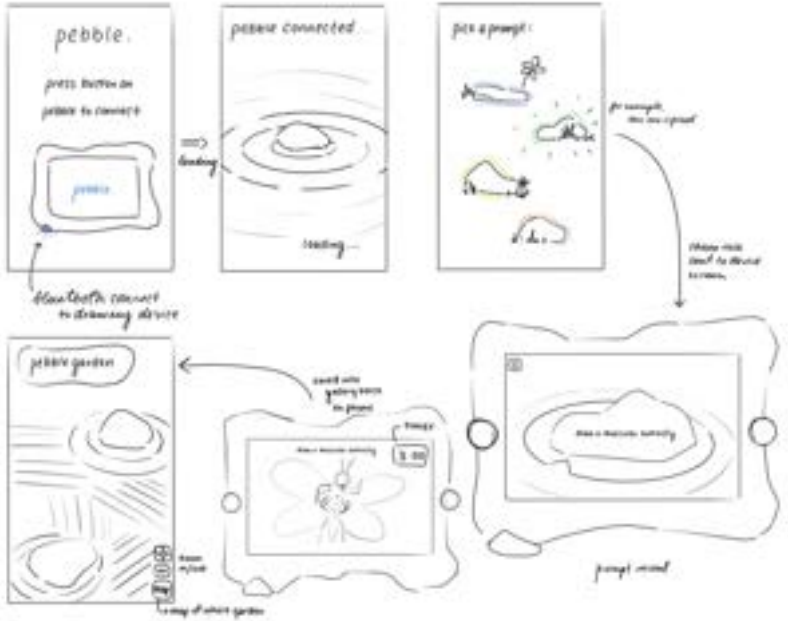
AR Sticker Collage: Cons



1. Harder to implement drawing element
2. Content moderation risk
3. Weaker fit for our needfinding and initial tasks
4. Heavily visual — some accessibility concerns

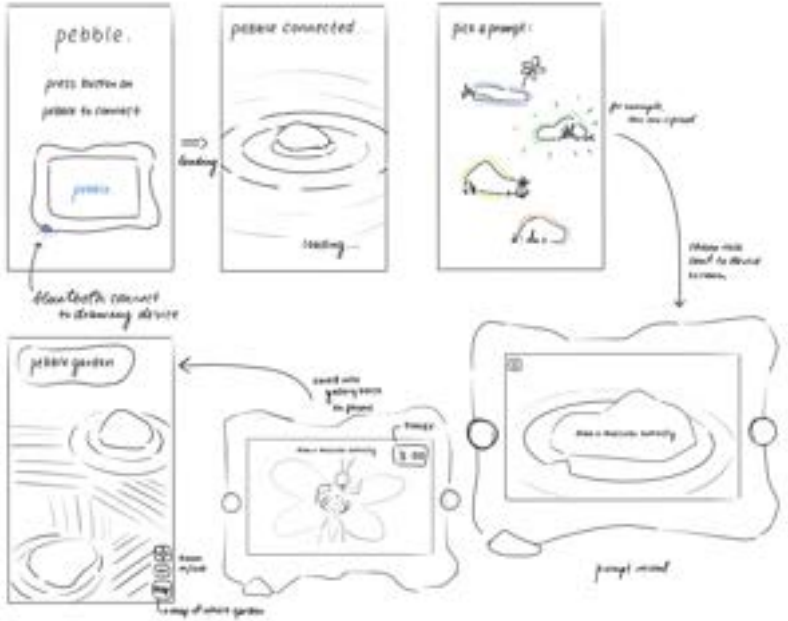
Physical Device: Pros

1. Matches our needfinding
2. Dedicated device -> fewer distractions
3. Timer and random prompts lower the pressure
4. Not mentally demanding



Physical Device: Cons

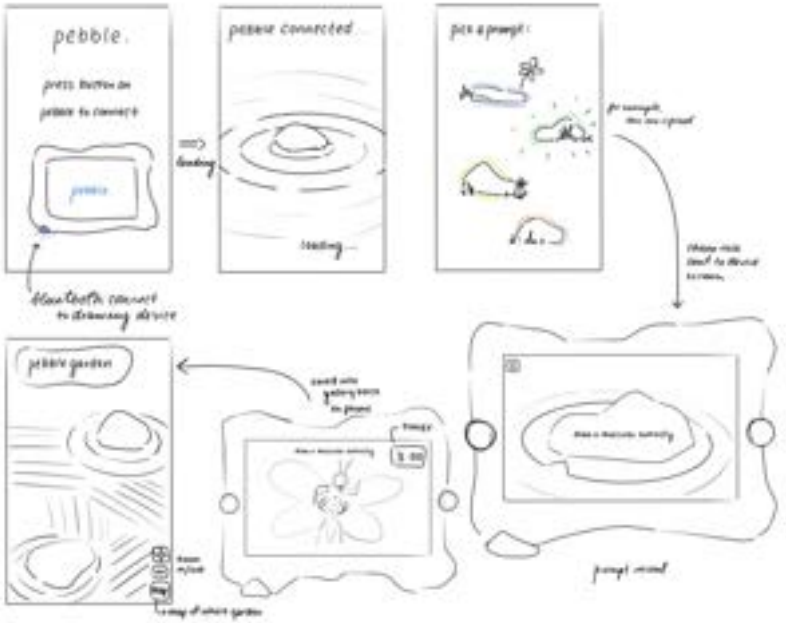
1. Must buy the device to start
2. Sharing requires friends to own one too
3. Easy to leave behind
4. Drawing only



What did we go with?

The physical pebble device.





Why pebble device over AR?

AR app: its cons hit the core loop

Drawing on a phone is awkward → drift photo-to-collage (assembling instead of making)

Hunting for real-world inspiration → effort and social awkwardness (opposite of low-stakes)

Without the AR layer → standard drawing-sharing app that maps poorly to our needfinding and tasks

device: its cons are peripheral

Missing social layer was addable → shared pebble garden and invite flow

Drawing-only scope fits the quick, deliberately silly sketch premise

Randomized prompts and a short timer lower the stakes

trade off we accepted

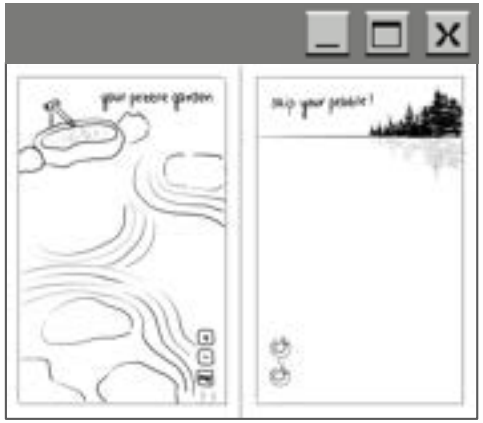
An extra device isn't technically required

Physical separation *is* the refuge → users off the device where their work lives

A second device for a protected break ✓

AR: richer capture and social features at the cost of the break

01 Lo-Fi Prototype Construction



Initially designed digitally

(Goodnotes, Procreate, etc.)

- Easy to share with each other
- Easy to edit without wasting paper







Then we printed everything out

- Physicality to prototype
- Easy to interact with

Lo-Fi Prototype

Task Flows

Initial Tasks

Simple:
- Take a quick creative break outside of work
- Pick a prompt and complete a drawing

Moderate:
- Make room for creativity on your own terms
- Reach mile-”stones”

Complex:
- Keep your creative practice alive through a busy stretch
- Invite a friend to the platform

Tasks We Added

Simple:
- Connect a device
- Keep or toss the drawing

Moderate:
- View gallery of previous work
- Share a drawing with friends

Complex:
- Invite a friend to a shared gallery
- Set a personal timer
- Edit personal gallery

Aggregate Task Flow: Connect Device and Complete a Drawing

Connect phone to device

Pick a prompt

Do the drawing

Keep or toss

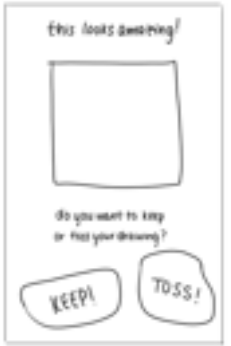




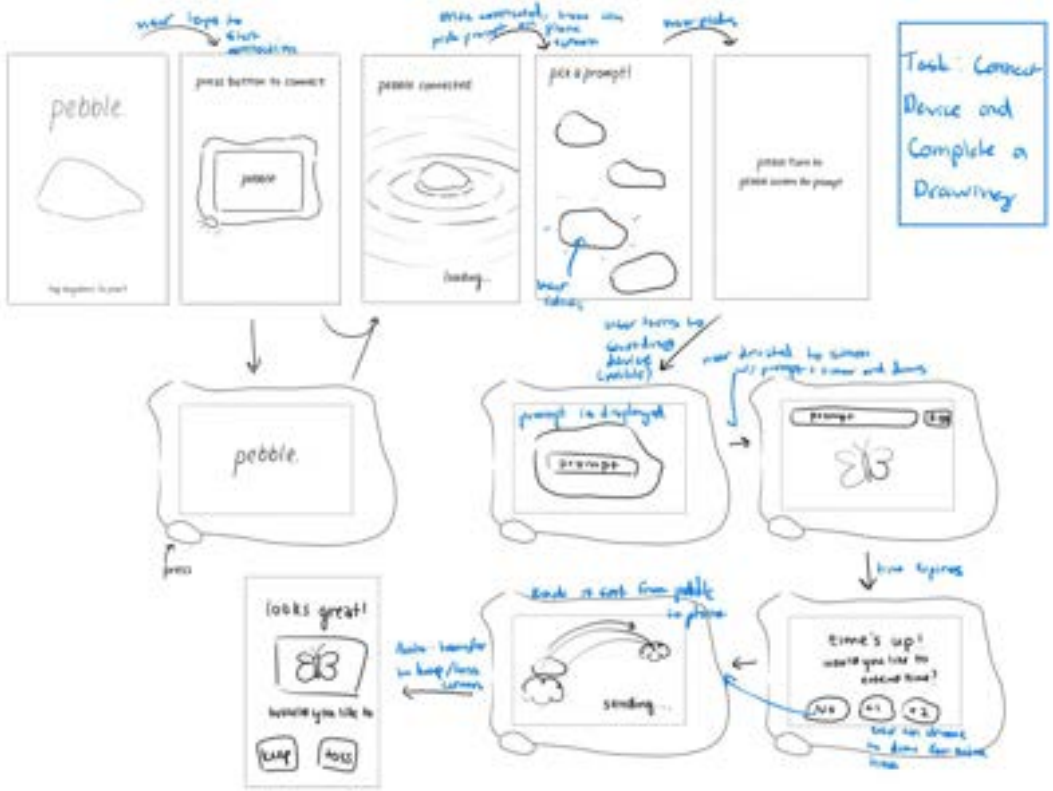


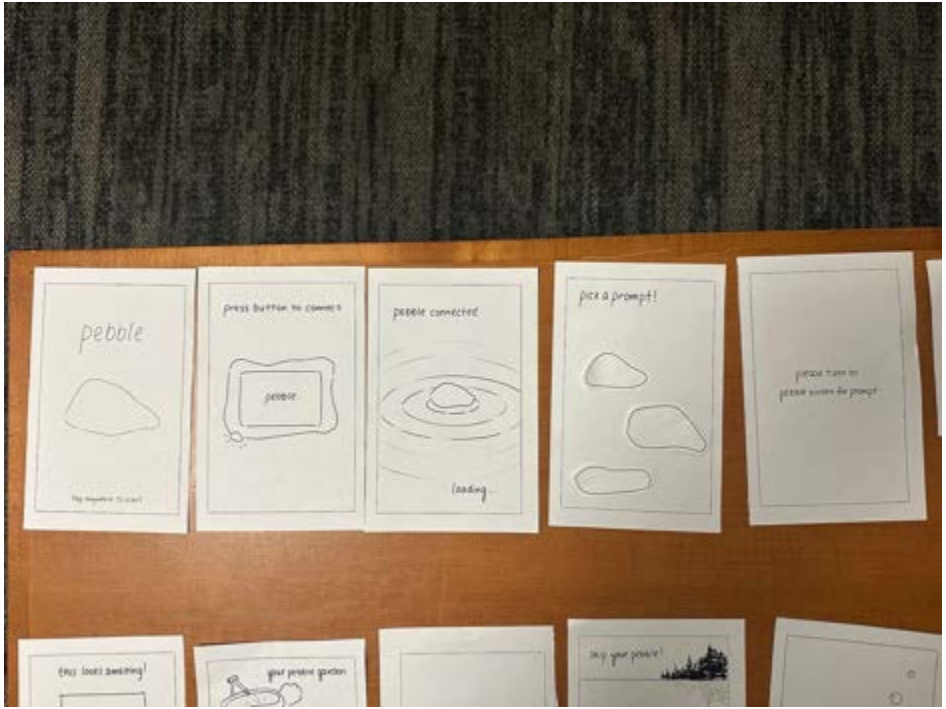






SLIDESGO


pebble
press button to connect
pebble connected
loading...
pick a prompt!
Task: Connect Device and Complete a Drawing
pebble
prompt is displayed
prompt
time's up!
looks great!
sending...


pebble
press button to connect
pebble
pebble connected
loading
pick a prompt!
please turn in
please answer the prompt
this looks awesome!
your prairie garden
in your prairie!

Aggregate Task Flow: Change the Timer Setting

Start at home page



Go to Settings



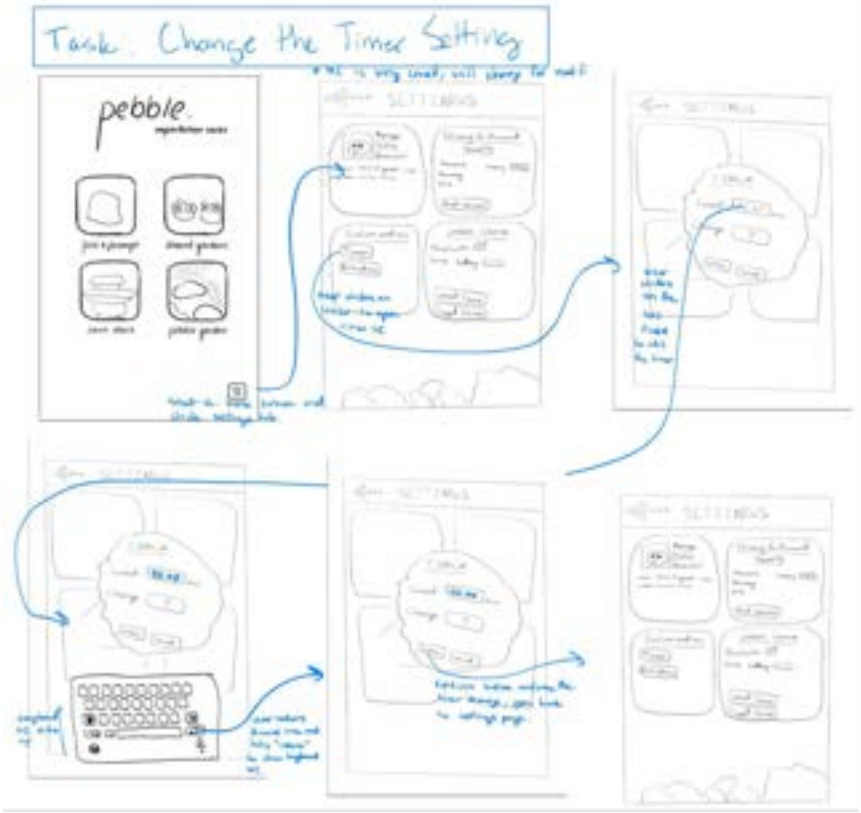
Select Timer Settings



Change the Time



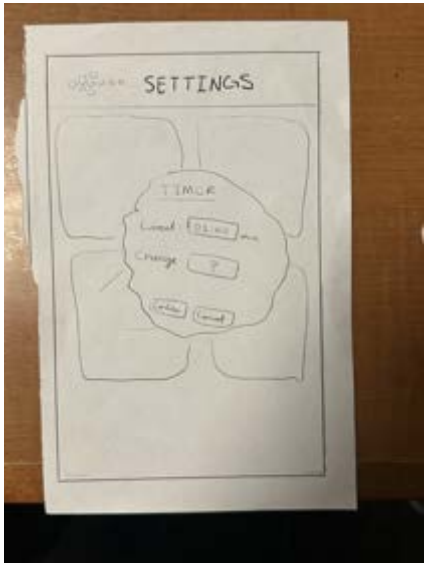
SLIDESGO


Task: Change the Timer Setting
pebble
SETTINGS

Changing the timer







Aggregate Task Flow: Create and Explore a Shared Garden

Invite someone to a shared gallery



Go to shared gallery



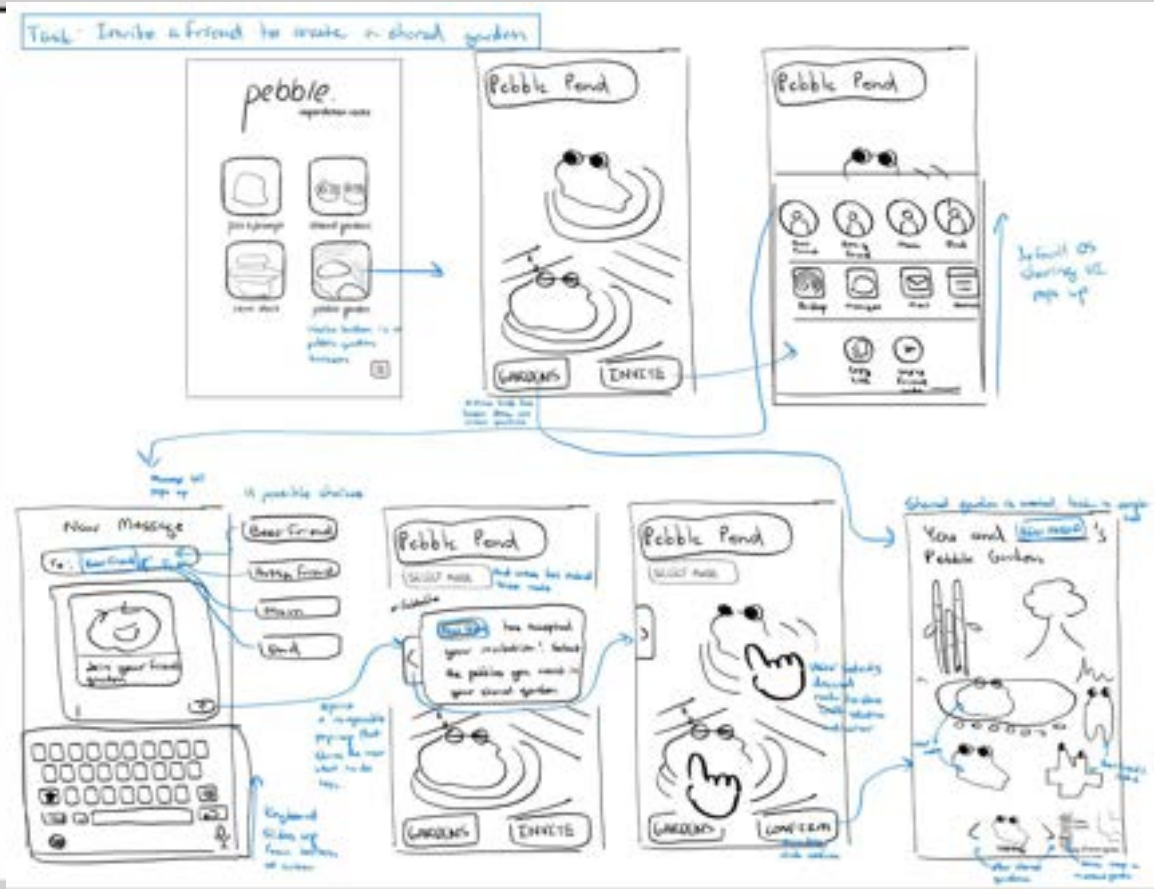
Check out a friend's drawing



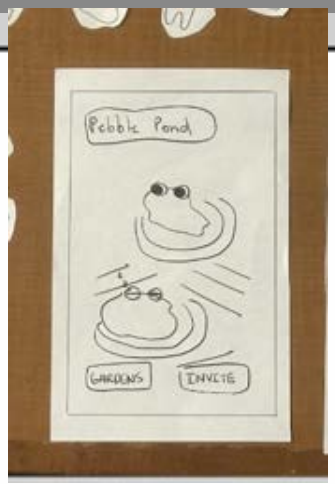
Send a message to a friend



SLIDESGO


Task: Invite a friend to create a shared garden
pebble
Pebble Pond
GARDENS
INVITE
Pebble Pond
New Message
Pebble Pond
GARDENS
INVITE
Pebble Pond
GARDENS
You and 's Pebble Gardens













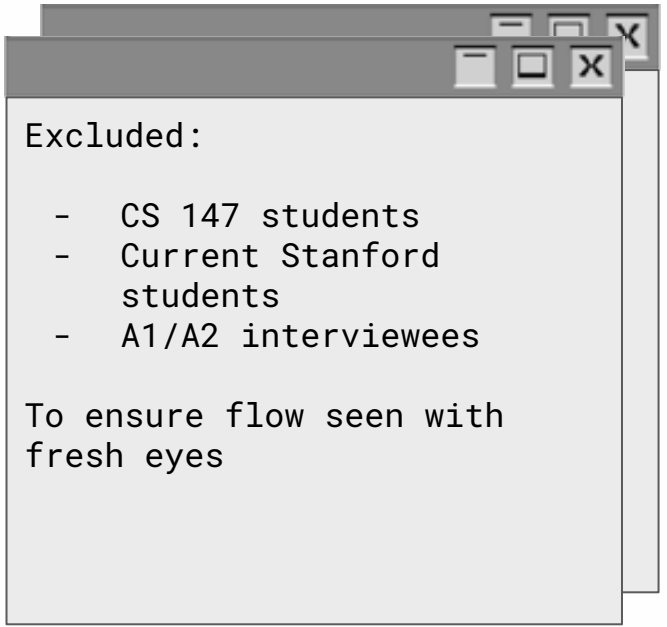


Shared Garden

Testing Methodology

Target Demographic and Participants

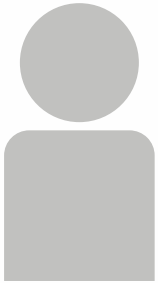
Busy students / working adults who want a creative outlet but say they lack the time



Participants

Participant 1

- Stanford alumni
- Current med student at Yale
- Digital art hobbyist



Why him?

The assumed demanding work associated with being a medical student + his hobby of drawing comics on the side.

Surveyed Tresidder during lunch and found him, and compensation was given as a free drink of choice.

Environment: sat at an outdoor table @Tresidder

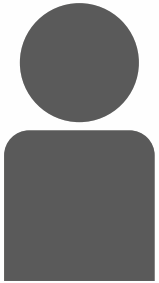
→ outdoor = quieter than inside,

→ enjoys taking breaks outside, said it aided in immersing into the mindset of ‘taking a break’.

Participants

Participant 2

- Stanford alumni
- Full time researcher at School of Medicine
- Enjoys microbiology art



Why him?

The long and unpredictable daily hours at work (sometimes staying until 10 pm at night to finish experiments) + his fun practice of creating art using microbes

Recruited around Stanford Bioengineering and School of Medicine, and offered a free meal as compensation.

Environment: table @ a café inside LKSC

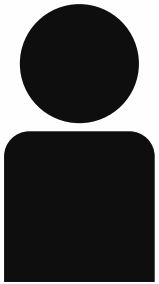
→ next to his lab, comes here often for his lunch breaks.

→ many medical students and employees also taking a break, so the ambient environment matched our testing conditions

Participants

Participant 3

- Stanford alumni
- ML engineer at a lab
- Previously acapella singer



Why him?

Since moving from student life to working life, he no longer has time for creative hobbies like drawing and a cappella.

Found through a friend. Compensated with Molly Tea from off campus.

Environment:@ their partner's dorm → visit often to hang out and spend time together

→ The dorm = relaxing and like a second home

→ Tested close to 10:30 pm, so lounge was quiet

→ Lounge simulated a place a busy person would want to unwind at after a long day

Environment and Apparatus







Member Roles

Brian

Facilitator:

- Introduced the test
- Provided context and guided participants through tasks
- Concluded test with questions about the participant's experience

Ryan

Notetaker:

- Sat across from the participant and recorded observations
- Noted participant responses during post-test questions

Cherie

Computer:

- Sat beside participant and operated the prototype in response to their actions
- Did not assist with tasks

Procedures

~ 20 minutes of participant's time

Contextualize user: class, product purpose, scope of test, consent form

Made clear: testing the prototype, not the user → think aloud of what you are feeling

Communicate what to expect from us:

- Complete a task, but won't tell how
- Transition to subsequent tasks upon completion
- User can interact with the prototype naturally; computer will update appropriately

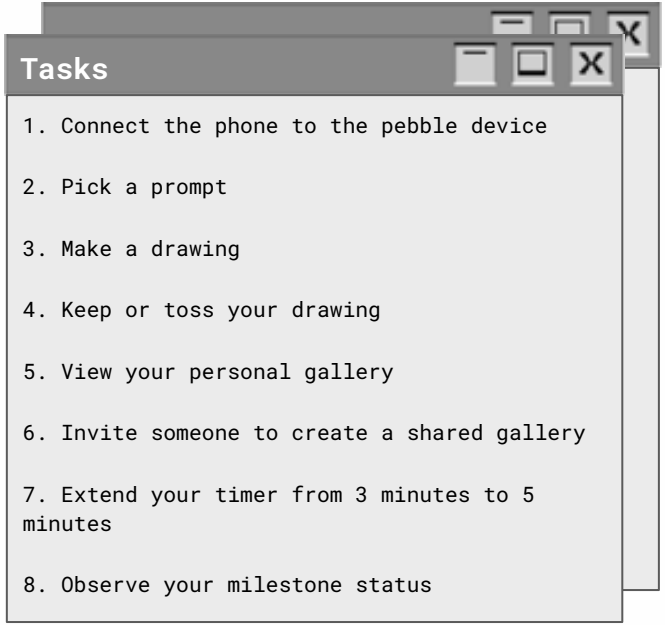
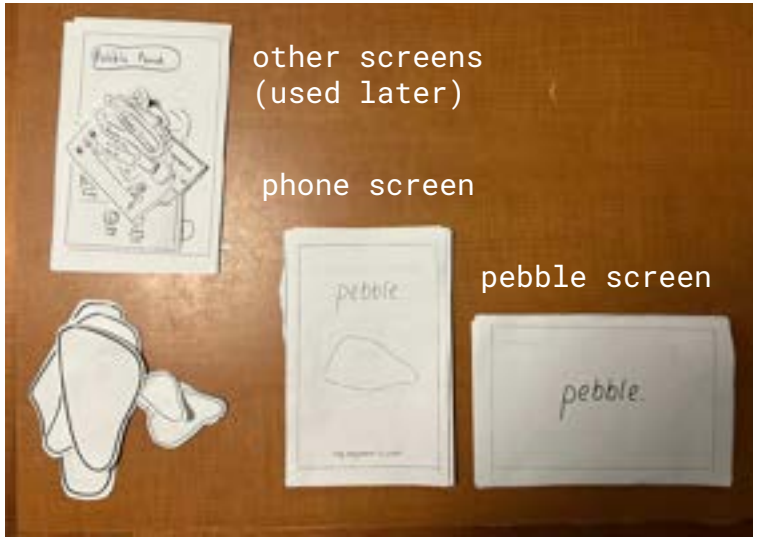


Photo of Setup

Initial Screen Setup



Test Setup

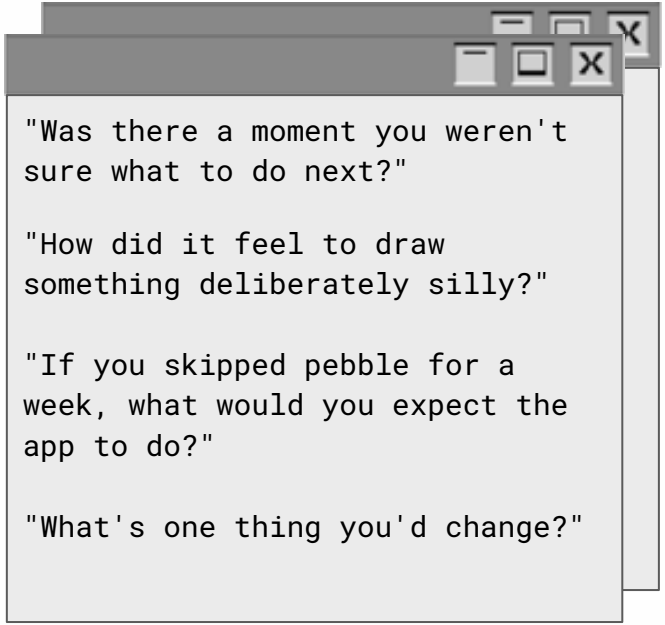


Procedures

Upon completion:

Follow-up debrief → ask questions (shown on the right)

Ask for ratings of key measurements (explored further in usability goals slides)



Usability Goals and Key Measurements

Goal 1: Pleasing

Pebble exists to give busy people a break, so a session should leave them calmer than it found them

Key measurement: change in self-rated relaxation (1 to 5) from before key task and after key task.

Target: no rating drops

- **Process data (observational):** signs of ease vs. signs of stress; think-aloud and debrief comments on how the session felt.
- **Bottom-line data (discrete):** relaxation rating before the test and after each key task

Usability Goals and Key Measurements

Goal 2: Discoverable

Many different tasks in prototype → participants should find each feature and explore intuitively

Key measurement: errors spent per task

Target: 0 on Task 1 to 4 (the core sequence), at most 1 on each other task

- **Process data (observational):** where they look or hover first; detours to the wrong screen; labels they misread; comments such as "where would that be?"
- **Bottom-line data (discrete):** # of assists, # errors per task, and completion (unaided, assisted, or failed)
- **Bottom-line data (time-based):** time spent on task

Testing Results

What worked:

- Participants completed the initial task flows with minimal confusion
- Silly prompts lowered the pressure on drawing
- Timer encouraged participants to continue drawing, even if they finished before the timer ended.
- The prospect of sharing with friends gave participants a bit more motivation to draw



What didn't work:

- The timer was also a source of stress → a way to end early if done
- Unclear how to connect the pebble device to the phone at the start
- Most friction came from navigation
- Missing some navigation buttons (e.g., home, back)





Usability Goal 1: Pleasing

Participants saw an increase in their self-rated relaxation after the overall testing experience

Participant 2 and Participant 3 reported verbally a small spike in stress after seeing that their drawing was timed. However, their relaxation scores did not drop.

- Observed amazement and positive remarks at the cairn stacking milestone tracker
- Many giggles and laughters during drawing
- The rock character drawings also led to increased laughter
- All participants decided to keep their drawings

Usability Goal 2: Discoverability

Most tasks were completed correctly.

Participant 2 and participant 3 both made an error at Task 1, which was connecting the phone to the pebble device.

Participant 3 struggled with finding out how to invite a friend to create a shared pebble garden.

- Even after the ‘connecting device’ screen instructed users to touch the device, participants regardless overlooked
- Participant 3 clicked repeatedly on the same button to comprehend how to get to shared pebble garden, to no avail
  - Led to longest time spent on a task, stuck in the home screen for ~30 seconds

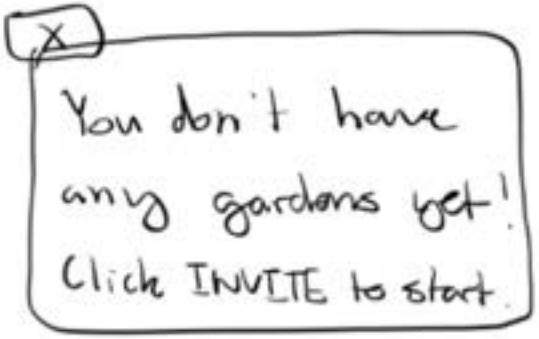
#1 Critical Incident

UI Mismatch with Inviting Friends to Shared Garden



Intended Button ->
<- No Invite Button





Prompted user to click “invite” on rock garden screen with no such “invite” button

Level 4 of confusion for user

Discussion

Implications:

Implication 1
Users shouldn't need to guess anything → more visible, intuitive UI that doesn't disrupt user flow

Implication 2
Having too much time can also add stress → give users the choice/freedom to end earlier

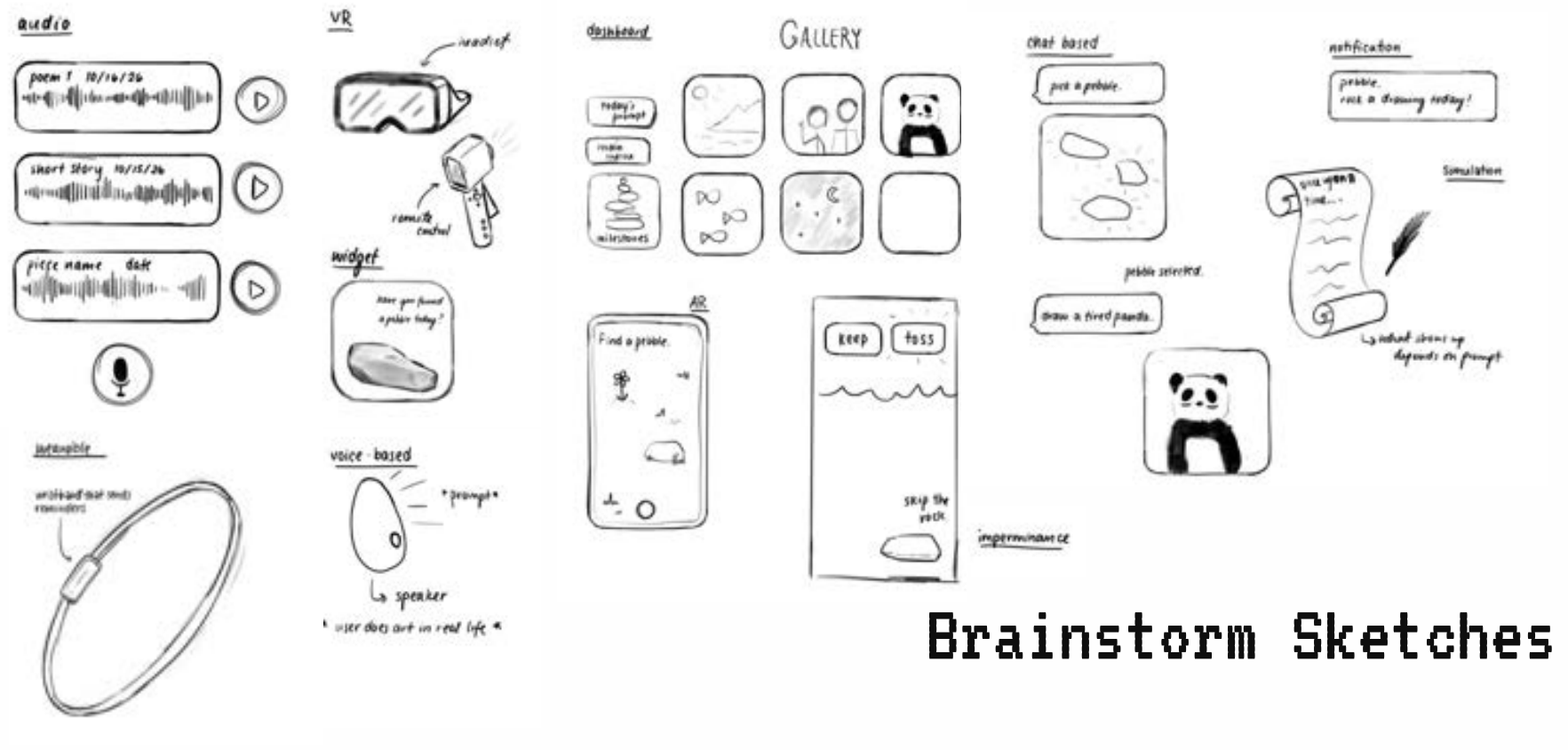
Implication 3
Simply having a second interface can create confusion despite efforts to make instructions clear

Implication 4
Even if drawings aren't perfect, people find satisfaction in keeping record of what they've drawn

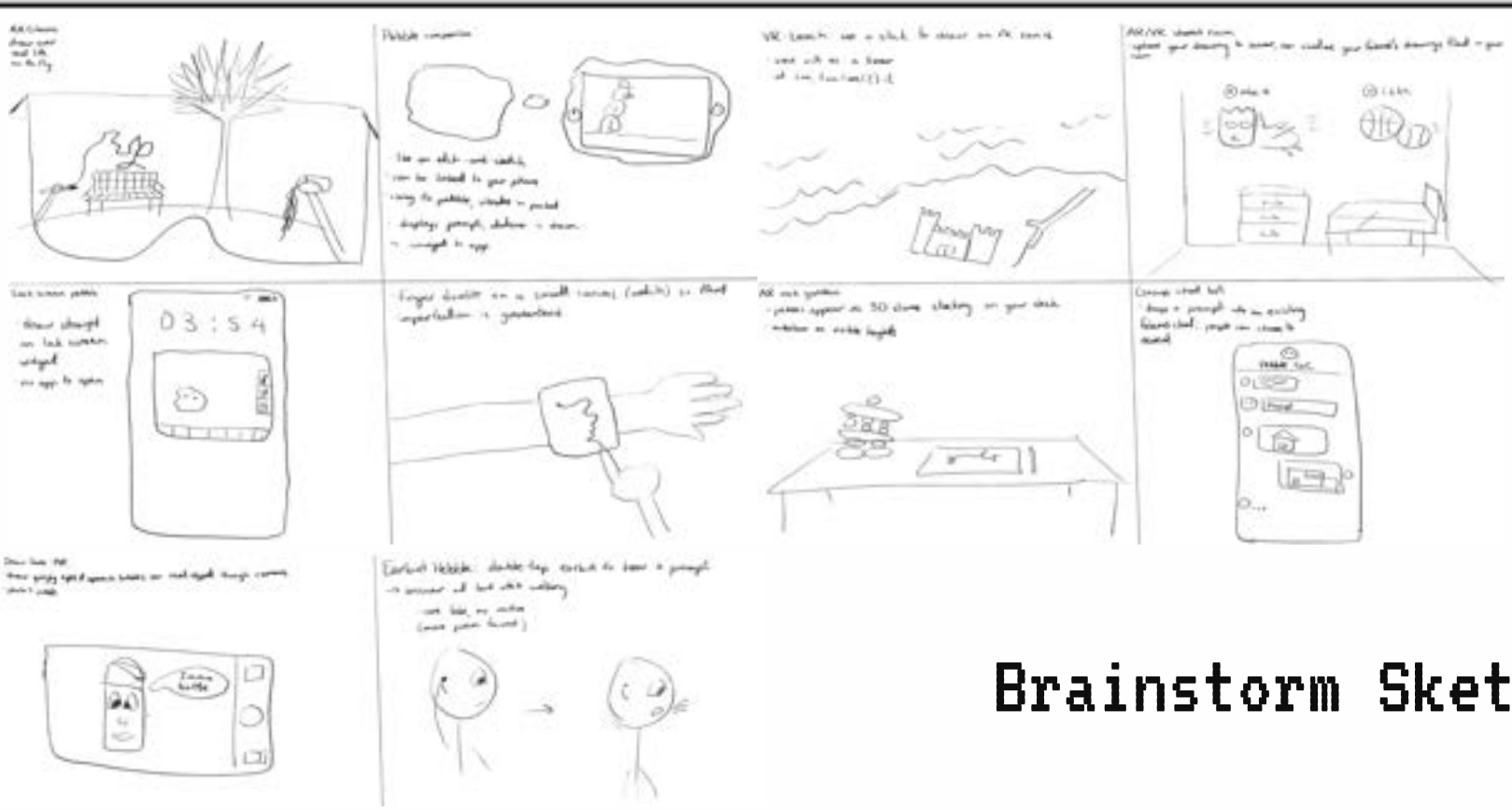
What Will We Change?

- During connection to pebble device, add explicit instructions to look at device screen
- Add consistent navigation buttons to prevent guessing
- Pivot access point of inviting friends into shared garden screen, where it appears more intuitive
- Add an end-early button during drawing
- Add small, interactive features (pop the bubbles, make the fountain flow in the garden) based on participant 1’s feedback

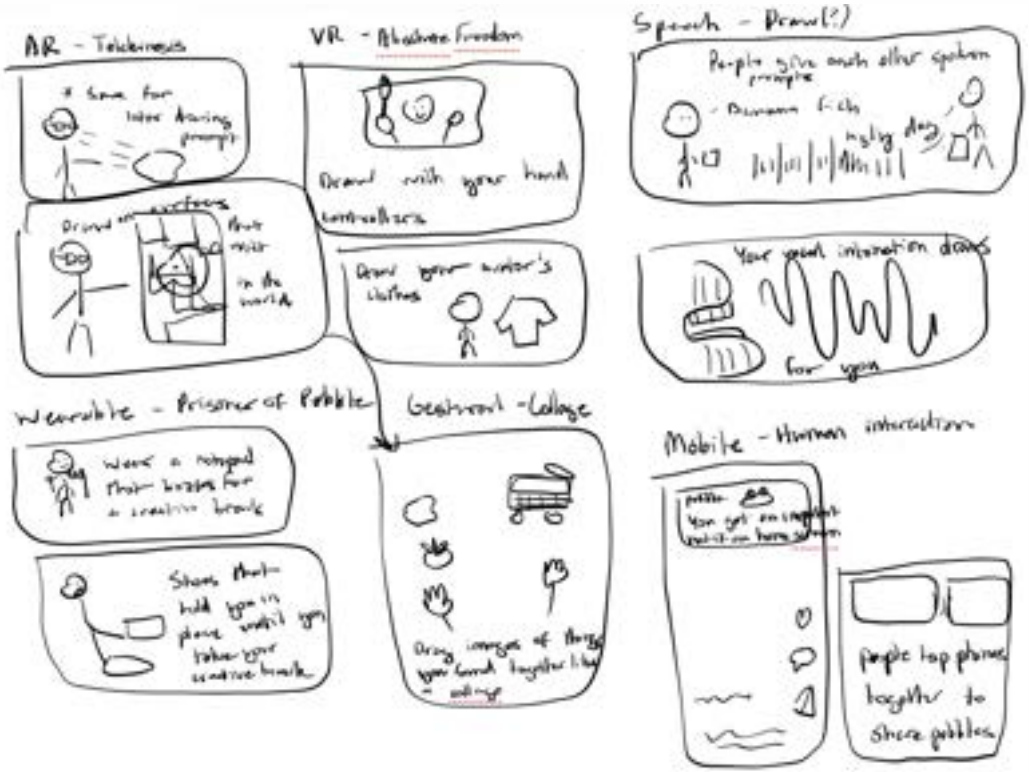
Appendix



Brainstorm Sketches

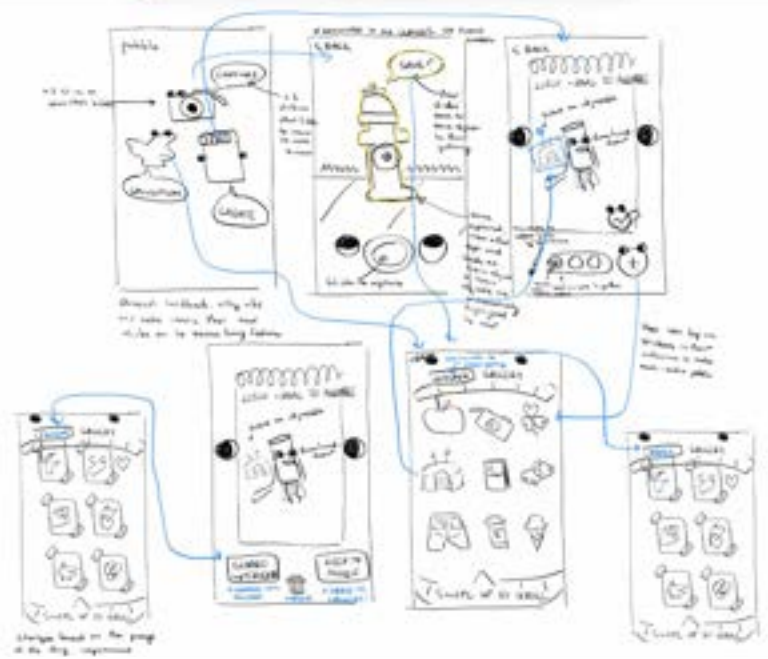


Brainstorm Sketches



Brainstorm Sketches

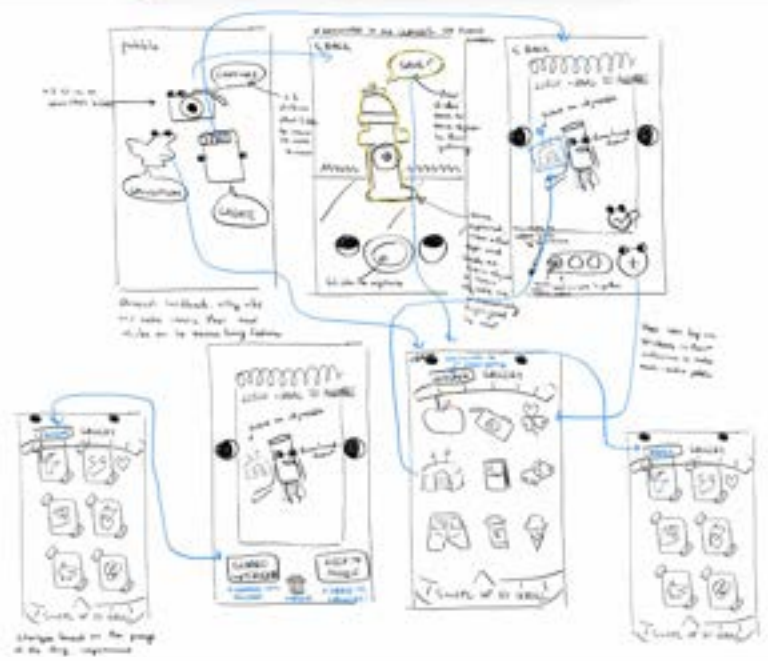
AR Sticker Collage



Pros

- interactive with the world (taking photos)
  - Augmented reality aspect allows for user intentionality with real world inspiration
- reusable icons/motifs – people can go back and reuse stickers they created previously
  - Stickers can be anything the user wants
- Allows user to share their drawings/collages with friends/engage in social interaction
- Gallery option allows users to look back on past collages/drawings; power/sense of accomplishment in milestones/scrolling through several collages
- Low barrier to entry: users can just use their camera app to take photos and make a collage

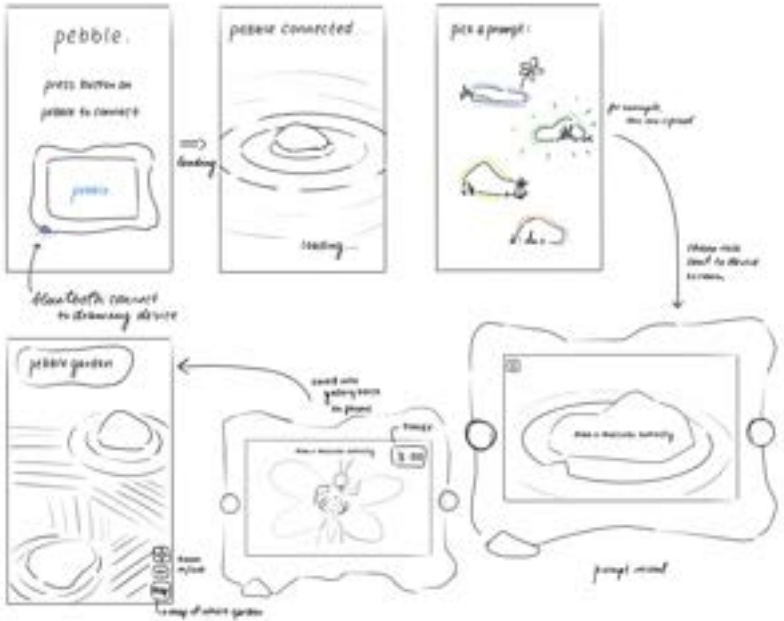
AR Sticker Collage



Cons

- harder to do actual drawing on the phone/maybe users will feel more incentivized to only do the photos -> collage which limits modes of creativity
- everything seems very visual, maybe accessibility concerns
- Stickers can be anything the user wants, so how do we make sure that there are no bad actors/bad actors get nullified before they can act badly
- (niche) User might have awkward interactions where they are inspired by a person's bag and have to ask them to take that picture
- possibility of confusion in the intuitiveness of the design if interface seems too "busy"/"crowded"
- Interface is not revolutionary; essentially a dedicated drawing-sharing app
- Doesn't fit nicely into needfinding or our tasks or our concept video

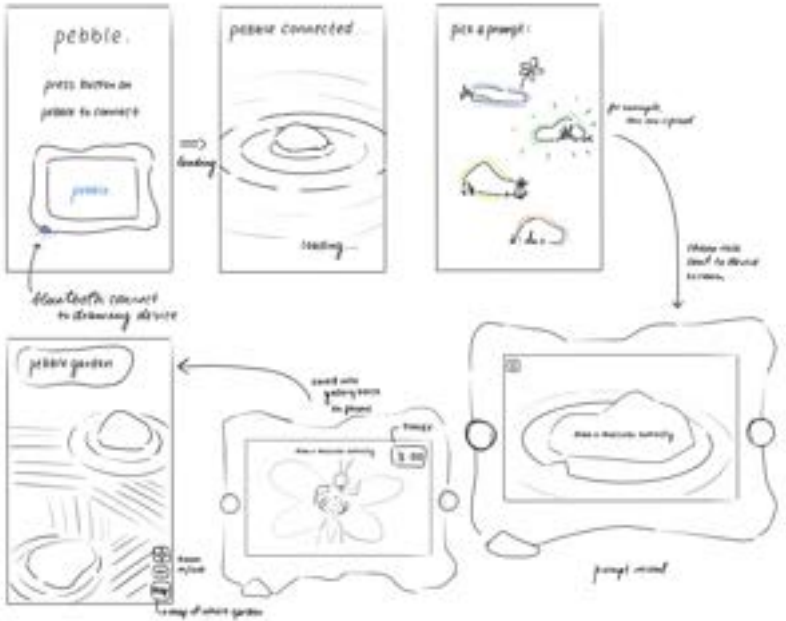
Physical Device



Pros

- Fits needfinding quite well/better than other realization (more of a refuge)
- Physical device stops user from getting distracted by phone while engaging in refuge
- Specifically uses timer and randomizes prompts which gives user lower stakes for what they draw
- There is no mentally ‘demanding’ tasks based on the simplicity of the design → all the user has to do is pick and draw, so there is less pressure to make anything demanding

Physical Device



Cons

- seems like it is limited to drawings only / unnatural to add other forms of art (ex. poetry…?)
- No social aspect, but definitely addable
- requires a whole other device as opposed to everything being central to one device
- Doesn't necessarily NEED a whole other device other than the goal of intentional mindfulness and a refuge from other responsibilities that might be present on a phone

Why pebble device over AR? (expanded in speaker notes)

AR app: its cons hit the core loop

Drawing on a phone is awkward → drift photo-to-collage (assembling instead of making)

Hunting for real-world inspiration → effort and social awkwardness (opposite of low-stakes)

Without the AR layer → standard drawing-sharing app that maps poorly to our needfinding and tasks

device: its cons are peripheral

Missing social layer was addable → shared pebble garden and invite flow

Drawing-only scope fits the quick, deliberately silly sketch premise

Randomized prompts and a short timer lower the stakes

trade off we accepted

An extra device isn't technically required

Physical separation *is* the refuge → users off the device where their work lives

A second device for a protected break ✓

AR: richer capture and social features at the cost of the break

Note on Technology Apparatus

A phone to record the interaction + a computer to take notes and read the script from.

Other than for these purposes, **no technologies** were used; all the prototypes were printed in **paper**. We used a double sided tape to incorporate interactivity with the user and to add dynamic features.

Testing Script

"Hi, thanks for helping us. We're Stanford students designing a phone app and complementary device called pebble, and today you'll try an early paper version of it. We're testing the app, not you. If anything is confusing, that's our fault, and it's exactly what we want to find."

[Hand over consent form.] "[**Notetaker**] will take notes and a few photos of your hands and the prototype, and no photos of your face unless you say it's okay. You can stop at any time."

"As you go, please think out loud: what you're looking at, what you expect to happen, and anything that surprises you. Tap the screen like a real phone and we'll change what you see. [**Computer**] can't talk, and I can't explain how things work, but I may ask what you're thinking."

"Imagine you just downloaded pebble. Your days are full of classes or work, and you got the app because you miss making things. We will ask you to complete a few tasks, but we won’t tell you how; a part of this testing relies on whether users can complete tasks intuitively."

Other Photos From Testing







Key Usability Measurements: Pleasing

"How relaxed do you feel currently?"

(5 is most relaxed, 1 is least relaxed)

Participant 1

Relaxation Rating

Before test: 3

Connecting devices: 4

After drawing: 5

Sharing with friends: 5

After test: 5

Participant 2

Relaxation Rating

Before test: 2

Connecting devices: 3

After drawing: 2

Sharing with friends: 3

After test: 4

Participant 3

Relaxation Rating

Before test: 3

Connecting devices: 4

After drawing: 4

Sharing with friends: 4

After test: 4

Key Usability Measurements: Discoverability

"How many errors on each task?"

Participant 1

1. 0
2. 0
3. 0
4. 0
5. 1
6. 1
7. 0
8. 1

Participant 2

1. 1
2. 0
3. 0
4. 0
5. 0
6. 2
7. 0
8. 0

Participant 3

1. 1
2. 0
3. 0
4. 0
5. 1
6. 5
7. 2
8. 0

Critical Incidents Log

Participant 1

Task	Critical Incident	Rating	Notes
Connect phone to pebble device	Participant initially pressed the pebble button on the phone screen instead of the physical button on the device.	**2**	Seemed unclear at first, but quickly realized after seeing that the computer didn't change anything.
Drawing	Continued drawing until the timer was done	**2**	Can be easily addressed with a small design change (i.e., "finished early" button)

Critical Incidents Log

Participant 2

Task	Critical Incident	Rating	Notes
Connect phone to pebble device	Didn't know the connection required pressing the physical device	**3**	
Drawing	** NOT AN INCIDENT ** but no way to indicate finishing drawing early other than verbal	**2**	Can be easily addressed with a small design change.
All	Said what he wanted to happen rather than clicking to make it happen	**2**	Likely because a person was operating the "computer." Will be resolved in digital prototype.
Picking Prompt	"Do I flip this?" when he selected a rock	**2**	The paper prototype's flippable elements invited a different interaction style than a digital screen would. A limitation of the paper prototype, not necessarily a usability issue with the design.
Looking at Personal Drawing	No back button on the drawing screen made it hard to navigate back after finishing.	**3**	

Critical Incidents Log

Participant 3

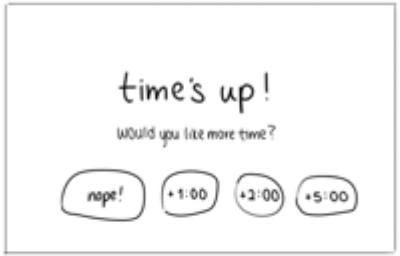
Task	Critical Incident	Rating	Notes
Connect Device and Complete a Drawing	Didn't click the button on the pebble to connect but did adequately click on the pebble's "screen."	**3**	
Complete a Drawing	Kept looking at timer and decided to draw more when he had a lot of time even when he perceived his drawing as done	**2**	
Change the Timer Setting	Intuitive but noted that the settings tab had really small font	**1**	Small text may hurt readability but this can be addressed with an easy design change.
Change the Timer Setting	Saw settings button as the same as the home button, hard to navigate the UI	**2**	
Create and Explore a Shared Garden	Didn't know where to find the invite button; "Click [Invite]" on the UI but no Invite button on the screen	**4**	

Task Flow: Complete a drawing

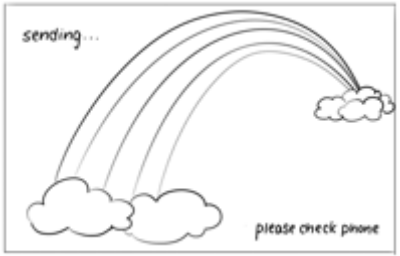
Draw on pebble device



Choose to extend timer

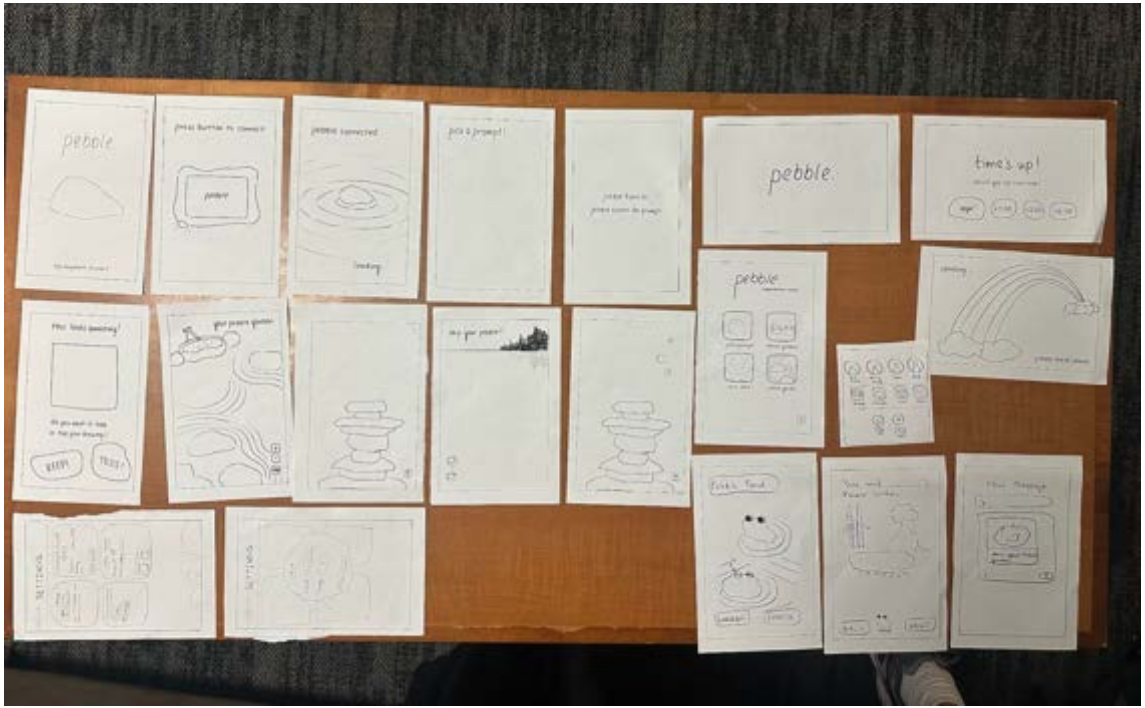


Finish drawing

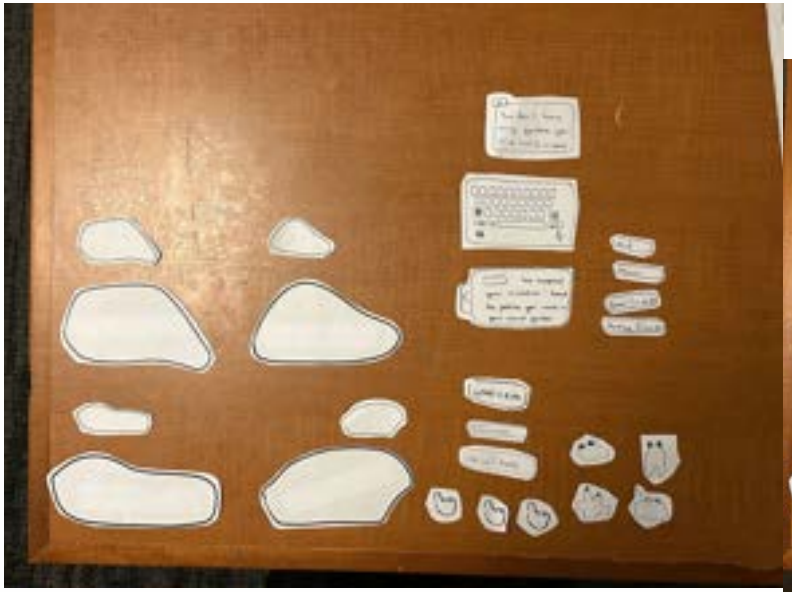


Paper Prototype Additional Photos

Bird's eye view, all screens



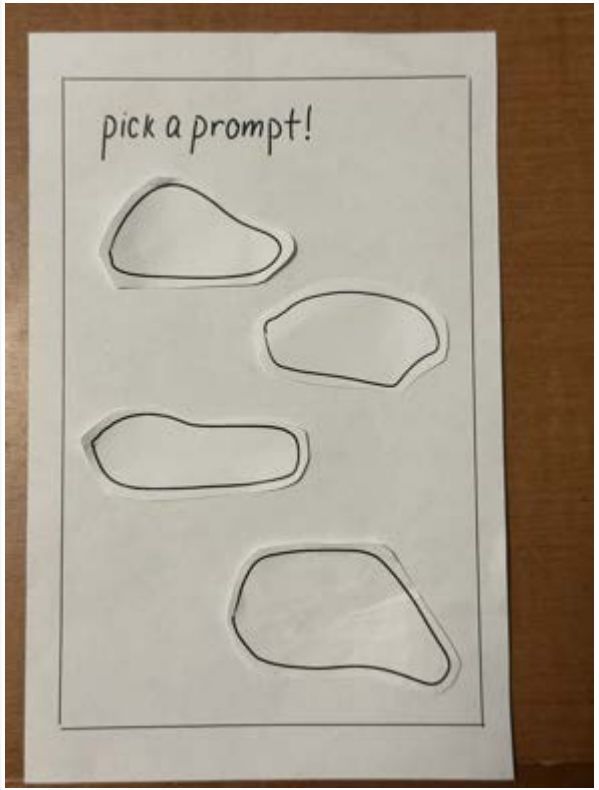
Bird's eye view, Interactable Objects





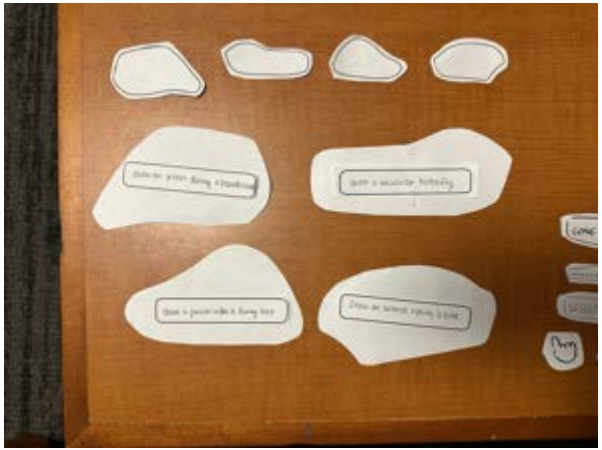


Picking a prompt to draw

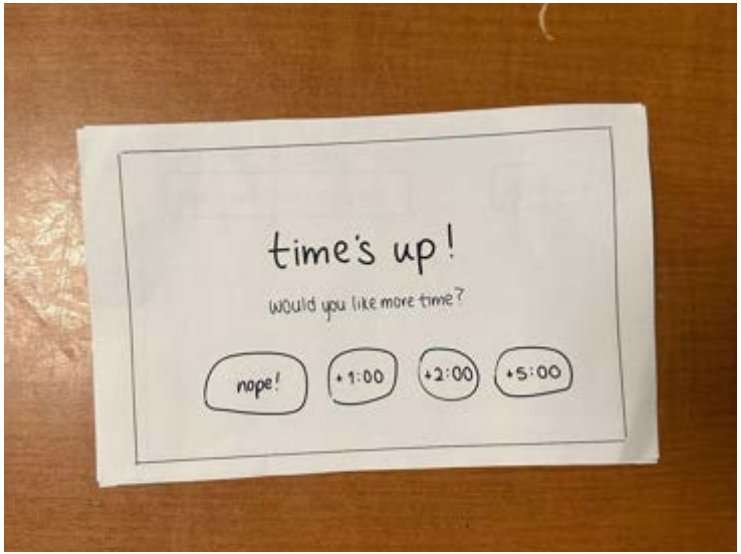


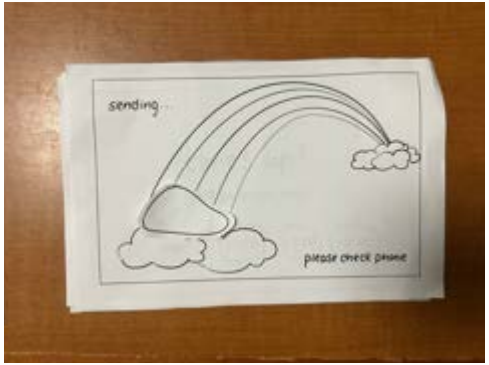






Adding time or sending drawing to phone from Pebble







Adding rock to garden





Skipping pebble when tossing





Cairn Stack (Pebble sinking when tossed)



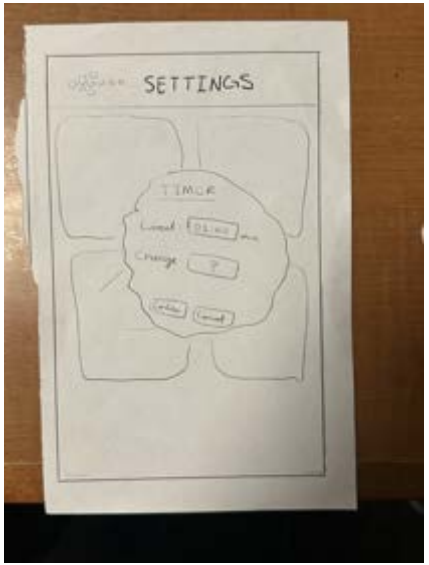




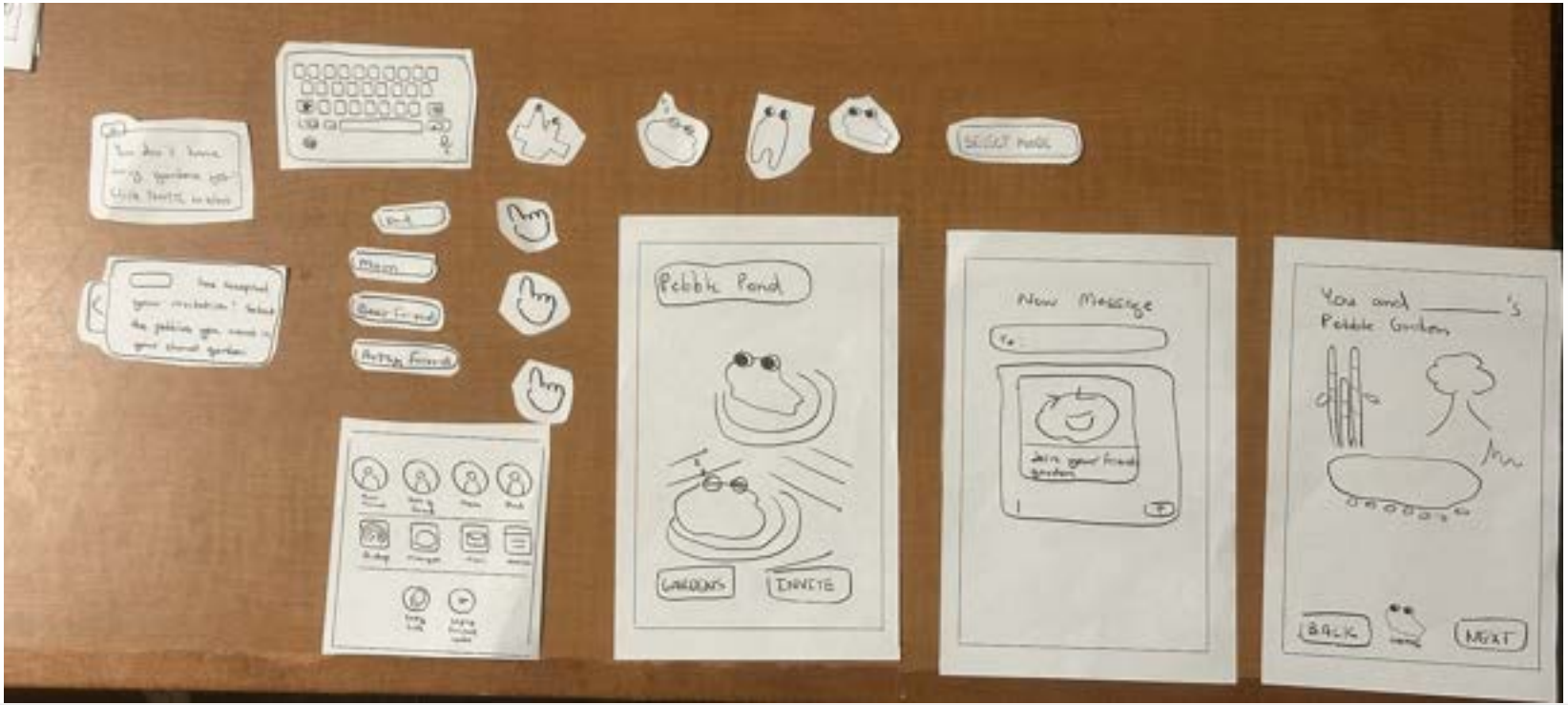
Changing the timer







Inviting Friend to Shared Garden



Inviting Friend to Shared Garden







Inviting Friend to Shared Garden



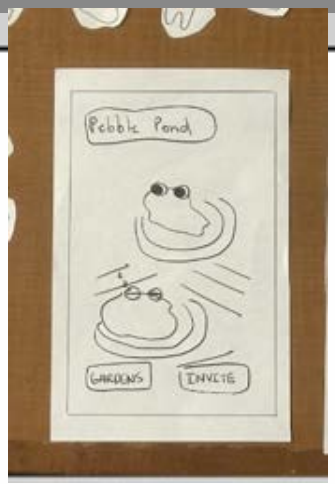




Inviting Friend to Shared Garden

















Full Flow, Shared Garden